Jun Huo

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Research Interests

My research focuses on flexible medical robots for surgery and wearable robots for rehabilitation and assistance, including supernumerary robotic limbs (SRLs), exoskeletons, and other bio-inspired robots. My work spans mechanism design, optimization, and learning-based control.

Education

- **Huazhong University of Science and Technology (HUST)** *Sep. 2018 – Dec. 2025*
  Ph.D. in Control Science and Engineering (Combined master's and doctoral program) Wuhan, China
  - Advisor: Prof. Jian Huang
  - Dissertation: Research on Multi-functional Universal Supernumerary Robotic Limb for Hemiplegic Patients Motion Assistance
- **Northeastern University (NEU)** *Oct. 2014 – Jun. 2018*
  B.E. in Mechanical Engineering Shenyang, China
  - Advisor: Prof. Zhiguo Lu
  - Thesis: Structural Design and Analysis of Hybrid Drive Humanoid Multi-Finger Dexterous Hand

Professional Experience

- **Research Associate, The Chinese University of Hong Kong (CUHK)** *Apr. 2026 – Mar. 2027*
  Mechanical and Automation Engineering (MAE) Hong Kong, China
  - Supervisor: Prof. Shing Shin Cheng

Honors and Awards

- **Best Paper Finalist Award** *Jul. 2024*
  2024 International Conference on Advanced Robotics and Mechatronics (ICARM)
  - Jun Huo, Xinyi Wang, Bo Yang, Qingyang Yan, Zhaofan Yuan, and Jian Huang*. **Active Compliance Variable Impedance Control of Lower Supernumerary Robotic Limb**.
- **The Sixth China Post-Graduate Student Robot Innovation Design Competition, National Second Prize** *Oct. 2024*
  China Association for Science and Technology
  - Jun Huo, Haoyuan Wang, Zhongzheng Fu, Xingjian Chen, Chengyao Li. A universal supernumerary robotic limb for motion function assistance
- **The Third China Post-Graduate Student Robot Innovation Design Competition, National Second Prize** *Aug. 2021*
  China Association for Science and Technology
  - Hongge Ru, Runzhe Zhang, Jun Huo, Cheng Chen, Mengshi Zhang. Pneumatic flexible hand for grip assistance in hemiplegic patients

Publications, Manuscripts, and Patents

J=Journal; C=Conference; P=Patent; S=Submitted manuscript; B=Book. * Corresponding author; # Equal contribution.

[J.1] Jun Huo, Jian Huang*, Jie Zuo, Bo Yang, Zhongzheng Fu, Xi Li, and Samer Mohammed. **Innovative Design of Multifunctional Supernumerary Robotic Limbs With Ellipsoid Workspace Optimization**. IEEE Transactions on Robotics, vol. 41, pp. 4699-4718, 2025. doi: 10.1109/TRO.2025.3588763.

[J.2] Jun Huo, Kehan Xu, Chengyao Li, Yu Cao, Jie Zuo, Xinxing Chen and Jian Huang*. **Variable Impedance Control for Floating-Base Supernumerary Robotic Leg in Walking Assistance**. IEEE Robotics and Automation Letters, vol. 10, no. 9, pp. 8698-8705, Sep. 2025. doi: 10.1109/LRA.2025.3588400.

[J.3] Jun Huo, Hongge Ru, Bo Yang, Xingjian Chen, Xi Li and Jian Huang*. **Air-Chamber-Based Soft Six-Axis Force/Torque Sensor for Human–Robot Interaction**. IEEE Transactions on Instrumentation and Measurement, vol. 73, pp. 1-12, 2024, Art no. 9501612. doi: 10.1109/TIM.2023.3338712.

[J.4] Jun Huo, Jian Huang*, Xikai Tu and Zhongzheng Fu. **Force sensorless admittance control of body weight support system**. Advanced Robotics, 35:7, 425-436, 2021. doi: 10.1080/01691864.2021.1875044.

[J.5] Jie Zuo#, Jun Huo#, Xiling Xiao, Yanzhao Zhang and Jian Huang*. **Human-robot Coordination Control for Sit-to-Stand Assistance in Hemiparetic Patients with Supernumerary Robotic Leg**. IEEE Transactions on Automation Science and Engineering, vol. 22, pp. 16591-16602, 2025. doi: 10.1109/TASE.2025.3578594.

[J.6] Zhongzheng Fu, Xinrun He, Enkai Wang, Jun Huo, Jian Huang* and Dongrui Wu. **Personalized Human Activity Recognition Based on Integrated Wearable Sensor and Transfer Learning**. Sensors. 2021, 21(3):885. doi: 10.3390/s21030885.

[J.7] Hongge Ru, Jian Huang*, Wenbin Chen, Caihua Xiong, Junzhe Wang and Jun Huo. **Design, modelling and identification of a fiber-reinforced bending pneumatic muscle**. Science China Information Sciences, vol. 62, no. 5, Art. no. 050213, 2019. doi: 10.1007/s11432-018-9709-x.

[J.8] Hongge Ru, Weijian Gao, Weixuan Ou, Xingyue Yang, Andong Li, Zhongzheng Fu, Jun Huo, Bo Yang, Yanzhao Zhang, Xiling Xiao, Zhaohui Yang, Jian Huang. **A Flexible Wearable Supernumerary Robotic Limb for Chronic Stroke Patients**. Journal of Visualized Experiments, 200, e65917, 2023. doi: 10.3791/65917.

[J.9] Kehan Xu, Jun Huo, Yize Zheng, Zixin Chi, Yu Cao, Zhaohui Yang and Jian Huang*. **Adaptive Gait Assistance for Foot Drop Rehabilitation Based on Uncertainty Fusion of Contralateral Limb Information**. IEEE Transactions on Neural Systems and Rehabilitation Engineering, vol. 34, pp. 2337-2348, 2026. doi: 10.1109/TNSRE.2026.3690647.

[J.10] Bo Yang, Zejia Zhang, Yu Cao, Jun Huo, Jie Zuo, Xiling Xiao, Wei Luo and Jian Huang*. **Gaze-Foot Hybrid Upper Limb Assistive Robotic System: An Innovative Approach to Aid Individuals with Limb Impairments**. IEEE Transactions on Biomedical Engineering, early access, pp. 1-13, 2026. doi: 10.1109/TBME.2026.3695684.

[J.11] Zhongzheng Fu, Haoyuan Wang, Yongkai Liao, Xinrun He, Xingjian Chen, Jun Huo and Jian Huang*. **Adaptive Fuzzy Residual Learning Framework for On-Skin Triboelectric Sensor Gesture Recognition**. IEEE Transactions on Fuzzy Systems, vol. 34, no. 8, pp. 2717-2730, Aug. 2026. doi: 10.1109/TFUZZ.2026.3695771.

[J.12] Wei Meng, Wenbo Liu, Xinyu Zhang, Jun Huo, Nan Xia, Jie Zuo*, Quan Liu and Jian Huang. **Long-Sequence Lower Limb Action Recognition via Action-Specific Attention and Logic Correction for Daily Rehabilitation**. IEEE Transactions on Neural Systems and Rehabilitation Engineering, vol. 34, pp. 3433-3444, 2026. doi: 10.1109/TNSRE.2026.3713524.

[J.13] Jun Huo, Rui Lei, Yuanyuan Yuan, Xiling Xiao, Xinxing Chen, Jie Zuo, Yu Cao and Jian Huang*. **Research on Robust Grasping of Wearable Supernumerary Robotic Limb by Integrating Gaze Guidance and Visual Perception**. Acta Automatica Sinica (Accepted).

[S.1] Yuanyuan Yuan, Yu Cao, Yifei Guo, Jun Huo and Jian Huang*. **Adaptive Soft-Constraint Barrier Lyapunov Functions for Safety-Critical Control with Application to Active Suspended Backpacks**. Automatica (Under revision, first revision).

[C.1] Jun Huo, Xinyi Wang, Bo Yang, Qingyang Yan, Zhaofan Yuan, and Jian Huang*. **Active Compliance Variable Impedance Control of Lower Supernumerary Robotic Limb**. 2024 International Conference on Advanced Robotics and Mechatronics (ICARM), Tokyo, Japan, 2024, pp. 19-24. doi: 10.1109/ICARM62033.2024.10715855.

[C.2] Jun Huo, Bo Yang, Hongge Ru, and Jian Huang*. **Parametric Design Optimization of a Universal Supernumerary Robotic Limb**. The 32nd 2021 International Symposium on Micro-NanoMechatronics and Human Science (MHS), Nagoya, Japan, 2021, pp. 1-6. doi: 10.1109/MHS53471.2021.9767156.

[C.3] Zhiguo Lu*, Jun Huo, Yuce Wang, Tongle Xin and Zhengbo Xie. **Design and simulation analysis of a lower limbs exoskeleton powered by hydraulic drive**. 2017 2nd International Conference on Advanced Robotics and Mechatronics (ICARM), Hefei and Tai'an, China, 2017, pp. 173-177. doi: 10.1109/ICARM.2017.8273155.

[C.4] Bo Yang, Jian Huang*, Menglin Sun, Jun Huo, Xiaolong Li and Caihua Xiong. **Head-free, Human Gaze-driven Assistive Robotic System for Reaching and Grasping**. 2021 40th Chinese Control Conference (CCC), Shanghai, China, 2021, pp. 4138-4143. doi: 10.23919/CCC52363.2021.9549800.

[C.5] Yu Cao, Jian Huang*, Zhangbo Huang, Xikai Tu, Hongge Ru, Cheng Chen and Jun Huo. **Dynamic Model of Exoskeleton Based on Pneumatic Muscle Actuators and Experiment Verification**. 2018 IEEE-RAS 18th International Conference on Humanoid Robots (Humanoids), Beijing, China, 2018, pp. 334-339. doi: 10.1109/HUMANOIDS.2018.8624914.

[P.1] Jian Huang, Jun Huo, Mengshi Zhang, Caihua Xiong, Xiling Xiao. (2021). **A human motion-assisted dual-use supernumerary robotic limb**. Chinese invention patent, Patent No. CN201910611936.0.

[P.2] Jian Huang, Jun Huo, Xikai Tu, Hongge Ru. (2021). **A kind of dynamic suspension weight support system**. Chinese invention patent, Patent No. CN201910694048.X.

[P.3] Jian Huang, Jun Huo, Bo Yang, Hongge Ru. (2022). **Parameter optimization method for supernumerary robotic limb based on workspace similarity**. Chinese invention patent, Patent No. CN202110609611.6.

[P.4] Jian Huang, Hongge Ru, Jun Huo, Xikai Tu, Xiao Li. (2022). **A pneumatic driven variable stiffness omnidirectional bending drive device**. Chinese invention patent, Patent No. CN202110423767.5.

[B.1] Jian Huang, Jie Zuo, Jun Huo, Yifei Guo. **Multi-functional Supernumerary Robotic Limb**. Science Press, Beijing, China. (Under preparation, funded by the National Science and Technology Academic Monograph Publication Fund of 2025).

Grants and Research Funding

- **National Science and Technology Academic Monograph Publication Fund of 2025** *Jan. 2026 – Dec. 2026*
  Co-PI, Support for the monograph (Science Press, Beijing, China)
  ◦ Jian Huang, Jie Zuo, Jun Huo, Yifei Guo. **Multi-functional Supernumerary Robotic Limb**.

Invited Talks

- **Multi-functional Universal Supernumerary Robotic Limb for Motion Assistance**
  ◦ Apr. 2025, Zhejiang University of Technology
  ◦ Jun. 2026, Hunan University

Academic Service

Journal Reviewer: IEEE Transactions on Robotics (T-RO), IEEE Transactions on Automation Science and Engineering (T-ASE), IEEE Robotics and Automation Letters (RA-L), Journal of Field Robotics (J-FR), IEEE Transactions on Neural Systems and Rehabilitation Engineering (T-NSRE), IEEE Transactions on Instrumentation and Measurement (T-IM), Advanced Intelligent Systems (AIS), Bioinspiration & Biomimetics (BB), Frontiers in Bioengineering and Biotechnology

Conference Reviewer: IEEE International Conference on Robotics and Automation (ICRA), Chinese Control Conference (CCC), IEEE International Conference on Cyborg and Bionic Systems (CBS), International Conference on Advanced Robotics and Mechatronics (ICARM)

Student Mentoring

- **HUST**
  - Rui Lei, undergraduate student, HUST, Research on Recognizing Limb Movement Intent Based on Gaze Guidance and EEG Decoding, Current: Master's student at HUST.
  - Wencan Tian, undergraduate student, Hunan University, Morphological self-optimization for multi-task human augmentation with a wearable robotic limb, Current: B.E. studies at Hunan University.
  - Chengyao Li, undergraduate student, HUST, Research on the Perception of External Limb Grasping with the Integration of Visual and Force Interaction; Current: Ph.D. student at HUST.
- **CUHK**
  - Yecheng Wang, undergraduate student, summer intern, Hysteresis Characterization and Compensation of a Cable-Driven Continuum Surgical Robot, Current: B.E. studies at the University of Leeds (UK).

References

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